QDARIA ZIPMINATOR PQC PLATFORM VALIDATION

Strategic Technology Assessment | Classification: Technical Due Diligence
Version: 2.0 | Research Depth: Comprehensive | Timeline: 2022-Present

EXECUTIVE SUMMARY

1. Hypothesis Validation Verdict

This report presents a comprehensive technical due diligence assessment of QDaria's Zipminator platform, validating its core three-pillar hypothesis. The analysis concludes that the hypothesis is **SUPPORTED WITH CAVEATS**. The strategic selection of NIST-standardized algorithms and the timing relative to powerful market drivers are strongly supported by evidence. The integration of a Quantum Random Number Generator (QRNG) is validated as a significant, high-assurance security enhancement. However, the reliance on the Mojo programming language for performance differentiation is currently speculative due to a lack of empirical evidence in cryptographic contexts and represents the most critical technical and execution risk.

- **Pillar 1: QRNG Integration with NIST PQC Standards**
  - **Confidence Score: A (HIGH)**
  - The value proposition of integrating a hardware QRNG is technically sound and directly addresses a well-documented class of implementation vulnerabilities related to randomness in cryptographic systems. Evidence from proof-of-concept integrations confirms feasibility with negligible performance overhead, and the commercial availability of certified QRNG hardware provides a clear path for production and regulatory acceptance.
- **Pillar 2: High-Performance Implementation in Mojo**
  - **Confidence Score: D (SPECULATIVE)**
  - This pillar is unsupported by direct evidence. No benchmarks or published research exist demonstrating Mojo's suitability for high-performance, constant-time cryptographic workloads. The claimed performance potential is an extrapolation based on proxy analysis of other languages and Mojo's marketing claims. The ability of the Mojo compiler to generate secure, side-channel-resistant code for complex operations like the Number Theoretic Transform (NTT) remains unproven and is a significant risk.

- **Pillar 3: Quantum Threat Landscape & Market Context**
    - **Confidence Score: A (HIGH)**
    - The selection of CRYSTALS-Kyber (ML-KEM) and CRYSTALS-Dilithium (ML-DSA) is unequivocally validated by their finalization as FIPS 203 and FIPS 204 by NIST and their inclusion in the NSA's CNSA 2.0 suite. The market opportunity is driven not by an imminent quantum threat but by powerful, time-sensitive regulatory and compliance mandates that create a clear and urgent demand for solutions built on these specific algorithms.

2. Strategic Implications

- **Primary Value Driver:** QDaria's most immediate and defensible value proposition is its perfect alignment with powerful regulatory tailwinds. The finalization of NIST PQC standards and the aggressive migration timelines set by the NSA's CNSA 2.0 mandate create a well-defined, compliance-driven market for which Zipminator's core algorithms are an exact fit.
- **Critical Risk—The Mojo Dependency:** The platform's reliance on the Mojo language presents a double-edged sword. If successful, it could yield a significant and defensible performance differentiator. However, the language's immaturity, the absence of a secure cryptographic ecosystem, and the unproven ability of its compiler to generate constant-time code represent a critical-path risk that could delay or derail the entire product roadmap. Security, not just speed, is the paramount concern.
- **Quantifiable Differentiator—The QRNG:** The integration of a hardware QRNG is a tangible, certifiable security enhancement. It provides a concrete defense against a known class of side-channel and fault-injection attacks that exploit weaknesses in randomness. This feature offers a clear, marketable point of differentiation against pure software-based PQC libraries, particularly in high-assurance markets.
- **Recommended Strategic Pivot:** The technology roadmap must be de-risked. It is strongly recommended that QDaria establish a parallel implementation track using a mature, high-performance language with a proven cryptographic ecosystem (e.g., Rust or C++ with AVX2 optimizations). This track would serve as a performance baseline, a security reference, and a commercial fallback. The Mojo implementation should be re-classified as a high-risk, high-reward research and development effort rather than the sole pillar of the go-to-market strategy.

3. Key Metrics Dashboard

Metric	Finding	Confidence
QRNG Security Advantage	Mitigates fault and	**A (HIGH)**

	side-channel attacks exploiting weak randomness. PoC integration shows negligible latency overhead ($<10^{-5}$) within a TLS handshake.[1]	
Mojo Performance Potential	*Extrapolated Projection:* Aims for parity with C++/AVX2, potentially 5-7x faster than reference C. Kyber-768 target: ≈ 0.034 ms total.[2]	**D (SPECULATIVE)**
Quantum Threat Timeline	Breaking Kyber-512 (NIST Level 1) requires quantum gate costs estimated at $\approx 2^{147}$, far exceeding current capabilities. Market driver is compliance, not imminent threat.[3]	**A (HIGH)**
Competitive Gap	No single commercial entity currently offers an integrated PQC (Kyber/Dilithium) + Hardware QRNG + High-Performance Language platform.	**B (MEDIUM)**

SECTION 1: QRNG INTEGRATION VALUE PROPOSITION

1.1 Entropy Requirements for NIST PQC Standards

The foundation of QDaria's first pillar rests on the specific requirements for randomness within the newly standardized post-quantum algorithms. The National Institute of Standards and Technology (NIST) finalized the standards for CRYSTALS-Kyber and CRYSTALS-Dilithium in August 2024, designating them as FIPS 203 (ML-KEM) and FIPS 204 (ML-DSA), respectively.[4] These documents form the definitive specifications for quantum-resistant key encapsulation and digital signatures for U.S. government use.

Both ML-KEM and ML-DSA rely fundamentally on the generation of random bytes for their security. These random bytes are used during key generation to create secret key material, during encapsulation to generate ephemeral secrets, and during signing to produce unique,

unpredictable nonces. For instance, the FIPS 204 standard for ML-DSA specifies a "hedged" signing procedure which internally uses a 256-bit random string, rnd. The standard mandates that this value "should be generated by an Approved Random Bit Generator (RBG)".[6] This requirement establishes a clear, compliance-driven need for a high-quality entropy source in any FIPS-validated implementation.

However, the standards define this requirement functionally rather than qualitatively. They mandate the *use* of an approved source, such as one compliant with NIST Special Publication 800-90, but do not explicitly differentiate between the statistical properties of a top-tier classical True Random Number Generator (TRNG) and a Quantum Random Number Generator (QRNG). Both types of devices can achieve the necessary certifications.[7] This creates a subtle but important distinction between baseline compliance and higher-assurance security. From a strict compliance-auditing perspective, a certified classical TRNG is sufficient. Therefore, the value proposition of a QRNG is not about meeting a standard that competitors cannot, but about addressing a more sophisticated threat model that goes beyond the letter of the FIPS documents. The argument for a QRNG hinges on providing a measurably higher quality of randomness that hardens the implementation against attacks that exploit the physical and statistical nature of the entropy source itself.

1.2 Security Advantages of QRNG vs. Classical Sources

The security advantage of integrating a QRNG is twofold, comprising a theoretical argument based on the nature of quantum physics and a practical argument rooted in mitigating a well-established class of implementation vulnerabilities.

Theoretically, QRNGs derive their randomness from intrinsically unpredictable quantum phenomena, such as the behavior of photons.[8] This provides an information-theoretic basis for the unpredictability of the entropy source. In contrast, classical TRNGs rely on complex but ultimately deterministic physical processes, while Pseudorandom Number Generators (PRNGs) are entirely algorithmic. A QRNG offers "true randomness" that is fundamentally non-reproducible and unbiased.[1]

The practical advantage is more direct and compelling. The history of cryptanalysis is replete with examples where the mathematical security of an algorithm was undermined by weaknesses in its implementation, particularly in the generation of random numbers. Research has demonstrated that lattice-based attacks can successfully recover private keys from Elliptic Curve Digital Signature Algorithm (ECDSA) implementations that used biased or predictable nonces.[9] These attacks, which have been used to compromise hundreds of cryptocurrency wallets, establish a critical precedent: **insufficient entropy is a proven, practical attack vector against deployed public-key cryptography.**

This threat directly translates to the PQC landscape. A significant body of research confirms that new PQC schemes, while mathematically resistant to quantum attacks, remain vulnerable to physical implementation attacks, such as side-channel analysis (SCA) and fault attacks.[10] These attacks exploit physical leakages like power consumption, timing variations, or

electromagnetic emissions to deduce secret key material. A predictable or biased random number generator can be a crucial enabler for such attacks. For example, research on deterministic lattice signatures shows that a single, well-timed fault injection can induce a nonce-reuse scenario, leading to full key recovery.[13]

By integrating a QRNG, the Zipminator platform aims to eliminate this entire class of vulnerabilities at their source. A QRNG provides the highest possible quality of entropy, making it significantly harder for an attacker to predict secret values or exploit statistical weaknesses in the random numbers used for key generation, masking, or nonce creation. This represents a form of defense-in-depth. In the nascent and rapidly evolving field of PQC implementation security, where new side-channel vulnerabilities are continuously being discovered [14], guaranteeing the integrity of the entropy source is a foundational architectural principle. It serves as a strategic hedge, hardening the platform not only against known randomness-based attacks but also against future, yet-to-be-discovered vulnerabilities that might exploit subtle statistical deviations in classical RNGs.

1.3 Implementation Evidence & Case Studies

The viability of integrating a QRNG into a PQC software stack is supported by recent experimental evidence. A proof-of-concept developed by researchers in the QSNP consortium, in collaboration with Nestlé, successfully integrated a hardware QRNG into a hybrid PQC protocol for Transport Layer Security (TLS).[1] This experiment utilized standard open-source cryptographic libraries, OpenSSL and the Open Quantum Safe (OQS) project's liboqs, demonstrating a clear and feasible integration path with the existing software ecosystem.

Crucially, this study also quantified the performance impact. The integration, which used an Entropy-as-a-Service (EaaS) model, showed that the QRNG itself added a negligible latency overhead of less than 10^{-5} to the TLS handshake. The total latency increase observed, under 30%, was primarily attributed to the larger key and signature sizes inherent to the PQC algorithms, not the process of sourcing entropy from the QRNG.[1] This provides strong evidence that integrating a QRNG is not a performance bottleneck and is practical for real-world, latency-sensitive applications.

The feasibility of this pillar is further bolstered by a mature commercial market for QRNG hardware. The table below summarizes the specifications for commercially available QRNGs from a leading vendor, ID Quantique, demonstrating their readiness for integration into various platforms.

Table 1: Commercial QRNG Hardware Specification Comparison

Vendor	Product/Model	Form Factor	Throughput	Key Certifications
ID Quantique	Quantis IDQ250C2	Chip	250 Kbps	NIST SP 800-90B
ID Quantique	Quantis IDQ6MC1	Chip	6 Mbps	NIST SP 800-90B, AEC-Q100

ID Quantique	Quantis IDQ20MC1	Chip	20 Mbps	NIST SP 800-90B, BSI AIS 31
ID Quantique	Quantis QRNG USB	USB Device	4 Mbps	METAS, CTL
ID Quantique	Quantis QRNG PCIe	PCIe Card	40 Mbps / 240 Mbps	METAS, CTL, BSI AIS 31

Source: Synthesized from [7]
As shown, QRNGs are available in a range of form factors, from miniaturized chips suitable for IoT and edge devices to high-throughput PCIe cards for servers. Their throughput rates, ranging from kilobits to hundreds of megabits per second, are more than sufficient to meet the entropy demands of PQC key generation. Most importantly, these products have achieved the stringent certifications required for use in cryptographic modules, including NIST SP 800-90B and the German BSI's AIS 31 standard, up to the highest PTG.3 level.[7] This pre-existing certification landscape significantly de-risks the regulatory and compliance pathway for a product like Zipminator.

1.4 Risk Assessment

The primary risks associated with the QRNG pillar are commercial and logistical rather than technical. The inclusion of a dedicated hardware component will invariably increase the bill of materials (BOM) cost and introduce supply chain dependencies. For cost-sensitive markets, this could be a disadvantage compared to pure software solutions. However, for QDaria's likely target markets—high-assurance sectors such as government, defense, and critical finance—the enhanced security posture and marketing value of "true quantum randomness" may justify the additional cost.
The risk of hardware failure is mitigated by features built into certified commercial QRNGs. Leading products include real-time health checks that continuously monitor the quantum entropy source to detect failures or potential tampering, ensuring the integrity of the output.[16] Therefore, with a robust supply chain strategy and a focus on appropriate market segments, the risks associated with this pillar are manageable.

SECTION 2: MOJO PERFORMANCE POTENTIAL

2.1 Lattice-Based PQC Computational Bottlenecks

To assess the potential performance advantage of an implementation in Mojo, it is first

necessary to identify the precise computational bottlenecks within the target algorithms, CRYSTALS-Kyber and CRYSTALS-Dilithium. Analysis and profiling studies of existing implementations reveal that the performance of these schemes is dominated by a few key operations centered around polynomial arithmetic over finite fields.

For **CRYSTALS-Kyber (ML-KEM)**, the computational workload is heavily concentrated in polynomial multiplication and the transformations required to perform it efficiently. Profiling studies indicate the following approximate breakdown of execution time for core operations:

- **Polynomial Multiplication (Poly_Mul):** $\approx 40\%$
- **Number Theoretic Transform (NTT):** $\approx 30\%$
- **Inverse NTT (INTT):** $\approx 30\%$ [17]

The NTT is a variant of the Fast Fourier Transform (FFT) adapted for polynomial rings, which reduces the complexity of polynomial multiplication from $O(n^2)$ to $O(n \log n)$.[18] However, the NTT itself is a complex algorithm involving recursive "butterfly" operations with intricate memory access patterns, making it a prime target for optimization.[20] Additionally, the process of sampling polynomial coefficients from a uniform or binomial distribution is a significant bottleneck, particularly in resource-constrained environments. This sampling often relies on rejection sampling from the output of a hash function like SHAKE-128, which can be inefficient in terms of latency and the number of random bytes required.[21]

For **CRYSTALS-Dilithium (ML-DSA)**, the bottlenecks are analogous. The core signing operation involves matrix-vector multiplication of polynomials, which is heavily reliant on the NTT for performance. The key bottlenecks are:

- **Matrix-Vector Multiplication:** This is the most expensive part of the signing process and is accelerated using the NTT. However, it still requires numerous forward and inverse NTTs, making the efficiency of the NTT implementation paramount.[18]
- **Polynomial Sampling:** Similar to Kyber, Dilithium requires sampling large matrices and vectors of polynomials from the output of SHAKE, which can be a latency bottleneck.[18]
- **Signature Rejection:** The Fiat-Shamir with Aborts paradigm used in Dilithium means that a signature attempt may be rejected if it falls outside certain statistical bounds, to avoid leaking information about the secret key. The entire signing process must then be repeated with a new nonce. Depending on the security level, this occurs on average 4 to 5 times per signature, directly multiplying the computational cost of the signing operation.[18]

The nature of these bottlenecks indicates that performance is not merely a function of raw floating-point operations. Instead, it is deeply tied to integer arithmetic, modular reductions, and memory efficiency—specifically, the ability to manage cache hierarchies and memory bandwidth during the complex data permutations of the NTT. Any high-performance implementation must excel in these specific areas.

2.2 Performance Benchmarking Analysis

A critical finding of this assessment is the complete absence of direct, empirical evidence for

Mojo's performance in cryptographic contexts. Available literature on Mojo focuses on its speed advantages over Python in general scientific computing (e.g., N-body simulations) or its application in Large Language Model (LLM) frameworks.[23] These benchmarks are irrelevant to the specific challenges of constant-time, integer-based cryptographic computation. Therefore, to establish a performance envelope for Zipminator, a proxy analysis based on existing high-performance implementations in other languages is necessary. This analysis sets a clear, quantitative target that any Mojo implementation must meet or exceed to validate its claims of superiority.

Proxy 1: C++ with AVX2 Optimizations (The Performance Gold Standard)

Vectorized implementations using Advanced Vector Extensions (AVX2) on modern x86-64 CPUs represent the current state-of-the-art for high-performance PQC software. A comprehensive benchmarking study provides the following concrete performance metrics, which serve as the primary baseline for this assessment.

Table 2: Performance Benchmarks: Kyber & Dilithium (Reference C vs. AVX2 C++)

Algorithm	Operation	Reference (ms)	AVX2 (ms)	AVX2 Speedup
Kyber-512	Key Generation	0.035	0.007	5.00x
	Encapsulation	0.040	0.007	5.71x
	Decapsulation	0.052	0.008	6.50x
Kyber-768	Key Generation	0.058	0.011	5.27x
	Encapsulation	0.063	0.011	5.73x
	Decapsulation	0.080	0.012	6.67x
Kyber-1024	Key Generation	0.089	0.015	5.93x
	Encapsulation	0.092	0.015	6.13x
	Decapsulation	0.113	0.017	6.65x
Dilithium-2	Key Generation	0.094	0.026	3.62x
	Signing	0.445	0.077	5.78x
	Verification	0.104	0.028	3.71x
Dilithium-3	Key Generation	0.167	0.045	3.71x
	Signing	0.665	0.120	5.54x
	Verification	0.160	0.045	3.56x
Dilithium-5	Key Generation	0.253	0.070	3.61x
	Signing	0.840	0.144	5.83x
	Verification	0.267	0.071	3.76x

Source: [2]

This data demonstrates that AVX2 optimizations yield a substantial **5-7x speedup for Kyber** and **4-6x speedup for Dilithium** over their reference C implementations. For Kyber-768 (NIST Level 3), the total time for a key exchange is reduced to just **0.034 ms**. This is the concrete performance target Zipminator must aim for.

Proxy 2: Rust

The Rust ecosystem has several PQC libraries supporting Kyber and Dilithium, and is

increasingly seen as a strong candidate for writing secure and performant systems code.25 Studies evaluating PQC in TLS have found that combinations like Dilithium2-Kyber512 implemented in Rust offer "rather good performance" in practice.27 Rust's key advantage is its compile-time memory and thread safety guarantees, which can prevent entire classes of bugs common in C/C++ cryptographic code.

Projected Mojo Performance Envelope

Based on Mojo's stated goals of combining Python's usability with C's performance and providing direct access to hardware features like SIMD, it is plausible that a highly optimized Mojo implementation could achieve performance comparable to the C++/AVX2 baseline. However, this remains a purely speculative projection (Confidence D) until supported by empirical benchmarks on cryptographic workloads.

2.3 Hardware Acceleration Feasibility

While QDaria's focus is on a software implementation, the potential for hardware acceleration is relevant context. GPU-based implementations demonstrate massive throughput improvements for batch processing, achieving speedups of over 100x compared to single-threaded CPU execution.[19] This is ideal for server-side applications handling many concurrent connections. FPGAs also offer a path to acceleration, with vendors like Lattice Semiconductor now offering "PQC-ready" devices with built-in crypto-agility to support algorithm updates.[29]

Mojo claims to have features for GPU integration. If these claims are valid, Mojo could potentially provide a more accessible path to GPU acceleration than low-level CUDA programming. However, this again runs into the critical challenge of ensuring security.

2.4 Constant-Time Implementation Challenges

The single greatest risk to the Mojo pillar is not its performance, but its ability to guarantee **constant-time execution**. This is a non-negotiable security requirement for cryptography. Side-channel attacks exploit data-dependent variations in the execution time or memory access patterns of an algorithm to leak secret information. To prevent this, cryptographic code must be written such that its execution path and timing are independent of any secret values it processes.

Achieving this is notoriously difficult, even in mature languages like C and Assembly, where cryptographers have decades of experience. High-performance features are often the primary source of timing leakages:

- **Compiler Optimizations:** A compiler, in its attempt to generate the fastest possible code, may introduce secret-dependent branches or look-up tables that create side-channels.
- **Variable-Time Instructions:** Certain CPU instructions, such as integer division (div),

can have execution times that vary based on the values of their operands. If these operands are secret, they create a direct timing channel.[15]
- **Cache Timing:** The time it takes to access memory depends on whether the data is in the CPU cache. If memory access patterns depend on secret data, this creates a cache-timing side-channel.

The Mojo language and its compiler are, from a security perspective, a black box. There is no body of research or documentation confirming that the Mojo compiler can be reliably instructed to produce constant-time machine code for complex algorithms like the NTT. The language's goal of providing seamless interoperability with the vast Python ecosystem is, from this perspective, a significant liability. Any call from a secure Mojo module into a standard Python library (e.g., for arbitrary-precision arithmetic) would inherit that library's side-channel vulnerabilities. To be secure, the entire cryptographic core of Zipminator would need to be written in a "pure," verifiable subset of Mojo, effectively negating the ecosystem advantage and requiring the development of a new, secure cryptographic library from scratch in an unproven language.

2.5 Critical Gaps & Research Needs

The validation of the Mojo pillar is blocked by a critical evidence gap. There are no publicly available benchmarks, security analyses, or case studies of Mojo being used for any constant-time cryptographic application. The claims of high performance are generic and have not been tested against the specific, complex bottlenecks of lattice cryptography. The security properties of the code generated by the Mojo compiler are completely unknown. Until QDaria can produce an implementation and subject it to rigorous, third-party performance benchmarking and side-channel analysis against the C++/AVX2 baseline, this pillar of their hypothesis must be considered speculative and high-risk.

SECTION 3: QUANTUM THREAT & MARKET TIMING

3.1 Quantum Attack Resource Requirements

The strategic context for PQC adoption is framed by the long-term threat of a Cryptographically Relevant Quantum Computer (CRQC). However, a sober analysis of the resources required to break the specific algorithms chosen by QDaria indicates that this threat is not imminent, and the primary market driver is regulatory action in anticipation of this future threat.

CRYSTALS-Kyber (ML-KEM):

NIST's own security analysis, current as of December 2023, provides the most authoritative

estimates for breaking Kyber. For Kyber-512, which corresponds to NIST Security Level 1 (equivalent to AES-128), the best-known classical attacks are considered more relevant than quantum attacks. The estimated quantum gate cost for a known attack on Kyber-512 is approximately 2^{147}.3 This is significantly higher than the 2^{128} operations required to break AES-256 via Grover's algorithm, and well above the 2^{64} operations needed for AES-128.31 This indicates that even the lowest security level of Kyber is robust against known quantum algorithms. Higher security levels (Kyber-768 and Kyber-1024) offer even greater security margins. While research into quantum circuits for Kyber's components, like the NTT, is ongoing, it is in a preliminary stage focused on resource estimation rather than demonstrating a practical attack.32

CRYSTALS-Dilithium (ML-DSA):

The security of Dilithium is based on the Module Learning With Errors (MLWE) and Module Short Integer Solution (MSIS) problems.33 These are well-studied lattice problems for which no efficient quantum solution is known. While specific qubit and gate count estimates for Dilithium are less detailed in the available literature, its security is categorized by NIST into the same levels as Kyber (Levels 2, 3, and 5, corresponding to ~128-bit, ~192-bit, and ~256-bit quantum security), implying a similar level of resilience.34

The analysis of these resource requirements leads to a crucial strategic conclusion: the market for PQC is not being driven by an immediate technical reality where existing encryption is actively being broken. Instead, the "quantum threat" serves as the narrative that underpins a powerful, compliance-driven market. Government agencies and standards bodies are not waiting for a CRQC to exist; they are acting now to mitigate a future risk. This creates a market dynamic where the opportunity is decoupled from the technical timeline of quantum hardware development. QDaria's success, therefore, depends on its ability to meet the immediate, tangible demand for PQC-compliant products generated by these regulatory actions, not on the actual arrival of a CRQC.

3.2 Quantum Computing Progress Indicators

Current state-of-the-art quantum computers have demonstrated logical qubit counts in the dozens, with IBM reporting a 48-logical-qubit device.[35] While this represents significant scientific progress, it remains orders of magnitude below the millions of physical qubits estimated to be required to break algorithms like RSA-2048.[36] Roadmaps from major players like IBM and Google project continued progress, but estimates for the arrival of a CRQC still vary widely, often placing it in the 2030s or beyond. This long-term horizon reinforces the conclusion that near-term market demand for PQC is a matter of policy and risk management, not immediate operational necessity.

3.3 QRNG Hardware Capabilities

As established in Section 1.3, the hardware required for the QRNG pillar of Zipminator's platform is mature, commercially available, and certified. A robust ecosystem of vendors, led by companies like ID Quantique, provides QRNGs in various form factors with sufficient throughput and the necessary regulatory approvals (e.g., NIST SP 800-90B, BSI AIS 31) to be integrated into cryptographic products.[7] The existence of this ecosystem de-risks the hardware component of QDaria's strategy, confirming its feasibility.

3.4 Regulatory & Market Drivers

The most powerful force shaping the PQC market is a set of concrete regulatory mandates and timelines from the U.S. government. These actions have transformed the PQC transition from a theoretical exercise into a time-sensitive business imperative for a large ecosystem of organizations.

NIST PQC Standardization and Migration Roadmap:

NIST's multi-year PQC standardization process culminated in the August 2024 publication of the first three final standards: FIPS 203 (ML-KEM/Kyber), FIPS 204 (ML-DSA/Dilithium), and FIPS 205 (SLH-DSA/SPHINCS+).37 This event serves as the official starting signal for the migration. Following the mandate in National Security Memorandum 10 (NSM-10), NIST is now developing official transition guidelines and timelines for the deprecation of vulnerable public-key algorithms.37 While the full transition across all industries is expected to take a decade or more, the publication of the standards creates immediate demand for compliant products and libraries.38

CNSA 2.0: The "Kingmaker" Mandate:

The Commercial National Security Algorithm Suite 2.0, published by the National Security Agency (NSA), is the single most significant market driver for QDaria's chosen technology stack. CNSA 2.0 explicitly specifies the algorithms required to protect U.S. National Security Systems (NSS). The suite includes:

- **Key Establishment:** ML-KEM-1024 (CRYSTALS-Kyber)
- **Digital Signatures:** ML-DSA-87 (CRYSTALS-Dilithium)
- **Symmetric Encryption:** AES-256
- **Hashing:** SHA-384/512 [39]

This selection effectively acts as a "kingmaker," creating a massive, well-funded, and non-negotiable market for products implementing Kyber and Dilithium. The entire ecosystem of government agencies, the defense industrial base, and critical infrastructure providers that interact with NSS will be required to adopt these specific algorithms. The NSA has published a clear and aggressive migration timeline, as summarized in the table below.

Table 3: CNSA 2.0 Migration Timeline

Date	NSA Mandate / Milestone
Now	Early deployment of CNSA 2.0 algorithms is encouraged as validated products become available.

Dec 31, 2025	No enforcement of CNSA 2.0 transition before this date.
Jan 1, 2027	All **new acquisitions** for National Security Systems must be CNSA 2.0 compliant.
Dec 31, 2030	All **fielded equipment and services** that cannot support CNSA 2.0 must be phased out.
Dec 31, 2031	**Full enforcement:** All cryptographic implementations in NSS must use CNSA 2.0 algorithms.
By 2035	All National Security Systems must be fully quantum-resistant per NSM-10.

Source: Synthesized from [39]

This timeline creates a powerful, predictable wave of demand. Furthermore, the NSA's guidance for NSS explicitly discourages the use of non-standardized or hybrid solutions for security purposes, preferring pure implementations of the CNSA 2.0 suite.[40] This gives a platform like Zipminator, which is built on the core CNSA 2.0 algorithms, a distinct advantage over hybrid approaches in this target market. QDaria's choice of algorithms is therefore not merely technically sound, but commercially astute, granting them a direct product-market fit with one of the most valuable and urgent early-adopter segments for post-quantum cryptography.

SECTION 4: COMPETITIVE POSITIONING & STRATEGIC RECOMMENDATIONS

4.1 Competitive Landscape Map

The competitive landscape for the Zipminator platform is fragmented, with no single commercial entity currently offering a directly comparable, fully integrated solution. The competition consists primarily of component providers and open-source projects, which potential customers would need to assemble themselves. This fragmentation represents QDaria's primary market opportunity.

Category A: PQC Implementation Platforms

This category is dominated by open-source libraries that provide implementations of the NIST PQC algorithms.

- **Open Quantum Safe (OQS) Project:** As the leading open-source C library for PQC, liboqs is a foundational project in the ecosystem.[41] It provides a wide array of

algorithms, including Kyber and Dilithium, and is used extensively for research and prototyping.[41] However, the OQS project explicitly warns against using its software in production environments, as it has not undergone the rigorous auditing required for high-assurance systems.[42] It functions more as a community enabler and a source of reference code than a direct competitor to a hardened commercial product.

- **Bouncy Castle:** A mature and widely used cryptographic library for Java and C#. It has integrated support for the NIST-standardized PQC algorithms, including Kyber and Dilithium.[43] For organizations operating within the Java/C# ecosystem, Bouncy Castle represents a strong and credible competitor for the software algorithm pillar.

Category B: QRNG Vendors with Cryptographic Integration

- **ID Quantique (IDQ):** As the market leader in QRNG hardware, IDQ is a key player.[16] They offer a range of certified hardware and promote quantum-safe solutions, but their focus appears to be on providing the entropy source as a component and on Quantum Key Distribution (QKD) systems, rather than offering an integrated PQC software platform like Zipminator.[16] IDQ is best viewed as a potential component supplier and strategic partner, but also a potential future competitor should they decide to move up the software stack.

Category C: High-Performance Cryptography Languages/Frameworks

This category is not composed of commercial competitors but represents the performance benchmarks Zipminator must surpass. It includes highly optimized C/C++ libraries utilizing AVX2/AVX-512 vector instructions (often the official NIST submission packages) and the burgeoning Rust cryptography ecosystem, which prioritizes both performance and memory safety.2

The following matrix summarizes the competitive positioning.

Table 4: Competitive Feature Matrix: Zipminator vs. Alternatives

Feature / Pillar	**QDaria Zipminator**	**Open Quantum Safe (liboqs)**	**Bouncy Castle**	**"DIY" (liboqs + IDQ)**
PQC Algorithm Support	Kyber (FIPS 203), Dilithium (FIPS 204)	Yes, broad support	Yes, in Java/C#	Yes, via liboqs
Entropy Source	Integrated Hardware QRNG	System PRNG / Pluggable	System PRNG	Hardware QRNG (manual integration)
Implementation Language	Mojo (unproven)	C (reference)	Java / C#	C
Performance Target	Exceeds AVX2 C++	Research/Prototyping	Standard Java/C#	Reference C
Constant-Time Guarantees	**Unknown / High-Risk**	Best-effort, not audited	Best-effort	User responsibility
Commercial Support	Yes (planned)	Community	Yes	None (component support only)
Integration Level	Fully integrated	Library	Library	Component-level

	platform			

This analysis reveals a clear "white space" in the market. No competitor offers a single, commercially supported, high-assurance platform that integrates certified hardware randomness with a high-performance PQC software stack. Customers seeking this level of security and performance today would be forced into a complex and risky do-it-yourself (DIY) integration project. This is QDaria's "integration moat" and its most powerful value proposition.

4.2 Differentiation Assessment

Unique Value Proposition: The unique value of Zipminator lies in the promised synergy of its three pillars. It is not just selling quantum-resistant algorithms; it is selling a holistic, high-assurance implementation of those algorithms. The platform's promise is to deliver NIST-standardized cryptography, seeded with the highest possible quality of entropy, and executed with best-in-class performance.
Defensibility: The platform's defensibility varies by pillar. The choice of algorithms is not defensible, as they are public standards. The reliance on Mojo is currently a weak point; until proven, it is more of a liability than a moat. The integration of a QRNG provides a more durable differentiator, as it requires hardware and systems engineering expertise. The strongest defensibility will ultimately come from the accumulated engineering effort in performance tuning, security hardening, and achieving formal certifications (like FIPS 140-3) for the entire integrated platform.
Market Positioning: Based on the analysis of market drivers and competitive gaps, the ideal initial market for Zipminator is the high-assurance sector. This includes:

- U.S. National Security Systems (NSS) and the surrounding defense industrial base, driven by the CNSA 2.0 mandate.
- Critical infrastructure operators (energy, finance, telecommunications).
- Other government agencies with long-term data confidentiality requirements.

These customers are most likely to value the enhanced security promise of the hardware QRNG, are less sensitive to the added BOM cost, and are driven by urgent compliance deadlines.

4.3 Strategic Recommendations (Prioritized)

Based on the evidence and analysis presented, the following strategic actions are recommended to maximize the probability of QDaria's success and mitigate critical risks.

- **HIGH PRIORITY: De-risk the Mojo Pillar.** The dependency on an unproven language for the core product is the single greatest threat to the venture. QDaria should immediately resource a parallel development track using a mature, performant language

with a robust cryptographic ecosystem, such as Rust or C++ with AVX2. This track will serve multiple critical functions:

1. It provides a reliable, low-risk path to a Minimum Viable Product (MVP).
2. It creates an internal performance and security baseline against which the Mojo implementation can be objectively measured.
3. It serves as a commercial fallback should the Mojo implementation fail to meet its performance or security targets in a timely manner.

- **HIGH PRIORITY: Focus Go-to-Market Strategy on CNSA 2.0.** The CNSA 2.0 mandate has created a time-sensitive, well-defined, and well-funded market for products based on Kyber and Dilithium. All product development, marketing, and sales efforts should be laser-focused on meeting the specific requirements of this market. This includes pursuing FIPS 140-3 validation and engaging with early adopters in the U.S. defense and intelligence communities.
- **HIGH PRIORITY: Conduct Rigorous Internal Benchmarking.** The core performance hypothesis for Mojo must be validated or invalidated as quickly as possible. QDaria must conduct internal, apples-to-apples benchmarks of its Mojo implementation of Kyber and Dilithium against the AVX2-optimized C++ reference implementations. These tests must measure performance in cycles-per-operation and, critically, must be accompanied by side-channel analysis (e.g., using tools like dudect) to test for constant-time properties.
- **MEDIUM PRIORITY: Formalize Security Audits and Partnerships.** Once a stable prototype exists, QDaria should engage a reputable third-party security firm to conduct a formal audit of the implementation, with a specific focus on side-channel vulnerabilities in the Mojo-generated code. Concurrently, the company should solidify its supply chain by formalizing a partnership with a leading QRNG vendor, such as ID Quantique, to ensure supply and explore co-marketing opportunities.
- **LOW PRIORITY/WATCH: Defer Broader Commercial Push.** Efforts to penetrate broader, more price-sensitive commercial markets should be deferred until the core technology is proven and a strong foothold has been established in the high-assurance sector. The unique value proposition of the integrated QRNG is strongest in markets where security assurance outweighs cost considerations.

4.4 Risk Mitigation

- **Technical Risks:** The primary technical risk is the failure of the Mojo implementation to deliver on its performance and security promises. The recommended parallel development track is the primary mitigation for this risk.
- **Market Timing Risks:** The market is driven by regulatory timelines. Delays in product development could cause QDaria to miss the critical procurement window opening in the 2025-2027 timeframe. The parallel development track also helps mitigate this by providing a faster path to an MVP.
- **Competitive Response:** A potential competitive risk is that a larger player (e.g., a major

cloud provider or a hardware security module vendor) could partner with a QRNG company to offer a similar integrated solution. QDaria's mitigation is speed and focus—by targeting the CNSA 2.0 market now, it can establish itself as the incumbent specialist before larger, slower-moving competitors can react.

SECTION 5: RESEARCH GAPS & FUTURE INVESTIGATION

5.1 Unanswered Questions

This investigation has revealed several critical areas where evidence is lacking, requiring further internal research and validation by QDaria.

- **Mojo's Constant-Time Guarantees:** This remains the most significant unknown. Does the Mojo language and its compiler toolchain provide the necessary low-level control and primitives to reliably generate constant-time machine code? Can it prevent the introduction of secret-dependent branches, variable-time instructions, or insecure memory access patterns during optimization? Without a positive answer, Mojo cannot be considered a secure language for cryptography.
- **Quantifiable Security Gain of QRNG:** While the theoretical and qualitative benefits of a QRNG are clear, there is a gap in research providing a hard, quantitative measure of its security improvement over a state-of-the-art classical TRNG (e.g., one compliant with NIST SP 800-90B) in the context of PQC. Quantifying the reduction in attack success rate or the increase in the number of traces required for a successful side-channel attack would provide a powerful metric.
- **Real-World Performance in Diverse Protocols:** The proof-of-concept showing low overhead for QRNG integration in TLS is promising.[1] However, performance data is needed for the full Zipminator platform across a wider range of protocols, such as IPSec, SSH, and secure boot processes, each of which has different latency and throughput characteristics.

5.2 Recommended Experimental Validation

To address these gaps and de-risk the technical roadmap, QDaria should prioritize the following internal experimental validation efforts:

- **Head-to-Head Performance and Security Benchmark:**
    1. Implement CRYSTALS-Kyber (specifically Kyber-768) in three languages: Mojo, C++ with AVX2 intrinsics, and Rust.

2. Execute all three implementations on the same, consistent hardware platform.
3. Measure performance in clock cycles per operation for KeyGen, Encaps, and Decaps. This provides a direct, hardware-independent comparison.
4. Perform non-profiled timing side-channel analysis on the core computational loops (especially the NTT) of the Mojo implementation using established tools (e.g., dudect) to test for constant-time properties. The results should be compared against the C++/AVX2 and Rust implementations, which have well-understood constant-time characteristics.

- **Integrated System Prototype Testbed:**
  1. Build an end-to-end prototype that integrates the selected commercial QRNG hardware with the software stack.
  2. Deploy this prototype in a realistic application scenario (e.g., as a TLS provider for a web server).
  3. Measure end-to-end performance metrics, including handshake latency, throughput, and CPU utilization under load.
  4. Conduct reliability and failure-mode testing on the QRNG hardware interface and its health-check mechanisms.

5.3 Ongoing Monitoring Strategy

The PQC landscape is dynamic. QDaria should establish a continuous monitoring strategy focused on the following areas:

- **Key Conferences and Journals:** Track proceedings from major cryptographic conferences (CRYPTO, EUROCRYPT, ASIACRYPT, PQCrypto, ACM CCS) and the IACR Transactions on Cryptographic Hardware and Embedded Systems (TCHES) for new research on lattice-based cryptanalysis and side-channel attacks.
- **Competitor Developments:** Monitor the activities of players in the PQC software space (Open Quantum Safe, Bouncy Castle) and QRNG hardware space (ID Quantique, Quintessence Labs) for new product releases or strategic shifts toward integrated solutions.
- **Regulatory Changes:** Closely follow updates from NIST regarding PQC migration guidance and from the NSA regarding CNSA 2.0 implementation details and timelines. Monitor similar activities from other national cybersecurity agencies like BSI (Germany) and ANSSI (France).
- **Mojo Language Evolution:** Track the development of the Mojo language and its compiler, specifically looking for the introduction of features or libraries relevant to low-level systems programming and security.

APPENDIX A: KEY RESEARCH PAPERS (Annotated Bibliography)

QRNG + PQC Integration

1. **Title:** Quantum Randomness Reinforces Post-Quantum Cryptography to Safeguard Large Enterprises in the Quantum-Safe Era
    - **Citation:** QSNP, 2024. [1]
    - **Tier:** 2
    - **Relevance:** This is the most critical paper supporting the feasibility of Pillar 1. It documents a proof-of-concept integrating a QRNG with PQC (using libOQS and OpenSSL) for TLS, finding negligible performance overhead from the QRNG itself.
    - **Confidence:** A (HIGH)
2. **Title:** Biased Nonce Sense: Lattice Attacks Against Weak ECDSA Signatures in Cryptocurrencies
    - **Citation:** De Santis et al., 2019. [9]
    - **Tier:** 1
    - **Relevance:** Establishes the critical precedent that weak randomness is a practical, exploitable vulnerability in public-key cryptography. While focused on ECDSA, its findings on lattice-based attacks to solve the hidden number problem directly motivate the need for high-quality entropy in all cryptographic systems, including PQC.
    - **Confidence:** A (HIGH)
3. **Title:** Quantis QRNG Product Certifications
    - **Citation:** ID Quantique, 2024. [7]
    - **Tier:** 3
    - **Relevance:** Provides definitive evidence of the commercial maturity and regulatory compliance of QRNG hardware. Documents key certifications including NIST SP 800-90B and BSI AIS 31, de-risking the hardware component of the Zipminator platform.
    - **Confidence:** A (HIGH)
4. **Title:** Differential Fault Attacks on Deterministic Lattice Signatures
    - **Citation:** Espitau et al., TCHES 2018. [13]
    - **Tier:** 1
    - **Relevance:** Directly demonstrates how fault attacks can create nonce-reuse scenarios in lattice-based signature schemes like Dilithium, leading to key recovery. This paper provides a concrete threat model that a high-quality entropy source like a QRNG helps to mitigate.
    - **Confidence:** A (HIGH)

PQC Performance & Optimization

5. **Title:** Performance Analysis and Industry Deployment of Post-Quantum Cryptography Algorithms
    - **Citation:** ResearchGate Preprint, 2025. [2]
    - **Tier:** 2
    - **Relevance:** This paper provides the most crucial quantitative data for Pillar 2. It contains detailed millisecond-level benchmarks for reference C vs. AVX2-optimized C++ implementations of all security levels of Kyber and Dilithium, establishing the "gold standard" performance target.
    - **Confidence:** A (HIGH)
6. **Title:** High-Performance Hardware Implementation of Lattice-Based Cryptography
    - **Citation:** Beckwith, NIST PQC Conference, 2022. [18]
    - **Tier:** 1
    - **Relevance:** Clearly identifies the primary computational bottlenecks in Dilithium: NTT-based matrix multiplication, polynomial sampling, and signature rejection. This analysis is essential for understanding where performance optimizations must be focused.
    - **Confidence:** A (HIGH)
7. **Title:** Performance analysis of the Kyber algorithm for post-quantum cryptography in HPC environments
    - **Citation:** CEUR Workshop Proceedings, 2025. [17]
    - **Tier:** 2
    - **Relevance:** Provides a percentage breakdown of execution time for Kyber's main operations (NTT, INTT, Poly_Mul), confirming that polynomial arithmetic accounts for nearly 100% of the computational cost and is the key area for optimization.
    - **Confidence:** B (MEDIUM)
8. **Title:** A Compact and High-Performance Hardware Architecture for CRYSTALS-Dilithium
    - **Citation:** Zhao et al., TCHES 2022. [46]
    - **Tier:** 1
    - **Relevance:** Details hardware optimization techniques for Dilithium, emphasizing the complexity of the operations and the need for specialized modules for NTT and sampling. This reinforces the difficulty of achieving high performance and the specific nature of the computational bottlenecks.
    - **Confidence:** A (HIGH)

Quantum Threat & Market Context

9. **Title:** Announcing the Commercial National Security Algorithm Suite 2.0
    - **Citation:** National Security Agency (NSA), 2022. [47]
    - **Tier:** 1
    - **Relevance:** The foundational document for the primary market driver. It officially

announces CNSA 2.0 and names the PQC algorithms (Kyber, Dilithium) required for protecting U.S. National Security Systems.
    - **Confidence:** A (HIGH)
10. **Title:** CNSA 2.0 Compliance Requirements, Algorithms & Timelines
    - **Citation:** SafeLogic, 2025. [39]
    - **Tier:** 3
    - **Relevance:** Provides a clear, consolidated table of the NSA's migration timeline for CNSA 2.0, outlining key dates for compliance and phase-out of legacy systems. This is critical for understanding the market window of opportunity.
    - **Confidence:** A (HIGH)
11. **Title:** FAQ for Kyber-512
    - **Citation:** NIST, 2023. [3]
    - **Tier:** 1
    - **Relevance:** Authoritative analysis from NIST on the quantum security of Kyber. It provides the key finding that breaking even the lowest security level of Kyber is likely harder for a quantum computer than classical attacks, contextualizing the quantum threat as a future, not immediate, problem.
    - **Confidence:** A (HIGH)
12. **Title:** NIST FIPS 203 & FIPS 204 Final Standards
    - **Citation:** NIST, 2024. [6]
    - **Tier:** 1
    - **Relevance:** These are the official standard documents for ML-KEM (Kyber) and ML-DSA (Dilithium). They are the ultimate source of truth for the algorithms' specifications and the basis for all compliant implementations.
    - **Confidence:** A (HIGH)

APPENDIX B: BENCHMARK DATA TABLES

(Included within the main body of the report as Tables 1, 2, 3, and 4)

APPENDIX C: METHODOLOGY NOTES

- **Search Queries Used:** "CRYSTALS-Kyber" AND "performance", "CRYSTALS-Dilithium" AND "bottleneck", "QRNG" AND "PQC" AND "integration", "Mojo language" AND "cryptography", "CNSA 2.0" AND "timeline", "quantum attack" AND "resource estimation" AND "Kyber".
- **Databases and Sources Consulted:** IACR ePrint Archive, NIST CSRC Publications, ACM Digital Library, IEEE Xplore, arXiv, Google Scholar, and targeted searches of vendor and open-source project websites.
- **Date of Research:** Research activities were concluded in the fourth quarter of 2025.

- **Limitations and Scope:** This report is based solely on publicly available information and the provided research materials. It does not include analysis of QDaria's proprietary source code, internal benchmarks, or financial data. The assessment of the Mojo programming language is limited by the profound lack of relevant public documentation and research concerning its use in secure, constant-time applications. All findings related to Mojo's performance and security potential should be treated as speculative until validated by empirical testing.

Works cited

1. Quantum Randomness Reinforces Post-Quantum Cryptography to ..., accessed October 27, 2025, https://qsnp.eu/quantum-randomness-reinforces-post-quantum-cryptography-to-safeguard-large-enterprises-in-the-quantum-safe-era/
2. (PDF) Performance Analysis and Industry Deployment of Post ..., accessed October 27, 2025, https://www.researchgate.net/publication/389917968_Performance_Analysis_and_Industry_Deployment_of_Post-Quantum_Cryptography_Algorithms
3. FAQ on Kyber512 - NIST Computer Security Resource Center, accessed October 27, 2025, https://csrc.nist.gov/csrc/media/Projects/post-quantum-cryptography/documents/faq/Kyber-512-FAQ.pdf
4. Post-Quantum Cryptography FIPS Approved | CSRC - NIST Computer Security Resource Center - National Institute of Standards and Technology, accessed October 27, 2025, https://csrc.nist.gov/news/2024/postquantum-cryptography-fips-approved
5. NIST FIPS 203, 204, 205 Finalized | PQC Algorithms | CSA - Cloud Security Alliance, accessed October 27, 2025, https://cloudsecurityalliance.org/blog/2024/08/15/nist-fips-203-204-and-205-finalized-an-important-step-towards-a-quantum-safe-future
6. FIPS 204, Module-Lattice-Based Digital Signature Standard | CSRC, accessed October 27, 2025, https://csrc.nist.gov/pubs/fips/204/final
7. Quantis QRNG Product Certifications - ID Quantique, accessed October 27, 2025, https://www.idquantique.com/random-number-generation/certifications/
8. What is a Quantum Random Number Generator (QRNG)? - Palo Alto Networks, accessed October 27, 2025, https://www.paloaltonetworks.com/cyberpedia/what-is-a-quantum-random-number-generator-qrng
9. Biased Nonce Sense: Lattice Attacks Against Weak ECDSA ..., accessed October 27, 2025, https://www.researchgate.net/publication/336437771_Biased_Nonce_Sense_Lattice_Attacks_Against_Weak_ECDSA_Signatures_in_Cryptocurrencies
10. A Look at Side Channel Attacks on Post-quantum Cryptography - SciELO México, accessed October 27, 2025, https://www.scielo.org.mx/scielo.php?script=sci_arttext&pid=S1405-5546202400

0401879
11. Side-Channel Resistance of Lattice-based Cryptographic Schemes - Aaltodoc, accessed October 27, 2025, https://aaltodoc.aalto.fi/bitstreams/260ab815-a338-47b6-ab4b-dbd5f22434d4/download
12. Side-Channel Attacks on Post-Quantum PKE/KEMs and Digital Signatures - DiVA, accessed October 27, 2025, https://kth.diva-portal.org/smash/get/diva2:2007310/FULLTEXT01.pdf
13. (PDF) Differential Fault Attacks on Deterministic Lattice Signatures - ResearchGate, accessed October 27, 2025, https://www.researchgate.net/publication/346707210_Differential_Fault_Attacks_on_Deterministic_Lattice_Signatures
14. Non-Profiled Higher-Order Side-Channel Attacks against Lattice-Based Post-Quantum Cryptography - ResearchGate, accessed October 27, 2025, https://www.researchgate.net/publication/396261106_Non-Profiled_Higher-Order_Side-Channel_Attacks_against_Lattice-Based_Post-Quantum_Cryptography
15. Divide and Surrender: Exploiting Variable Division Instruction Timing in HQC Key Recovery Attacks - USENIX, accessed October 27, 2025, https://www.usenix.org/system/files/usenixsecurity24-schroder.pdf
16. Quantum Random Number Generation (QRNG) - ID Quantique, accessed October 27, 2025, https://www.idquantique.com/random-number-generation/overview/
17. Performance Evaluation and Profiling of Kyber for ... - CEUR-WS.org, accessed October 27, 2025, https://ceur-ws.org/Vol-4055/icaiw_waai_9.pdf
18. High-Performance Hardware Implementation of CRYSTALS-Dilithium, accessed October 27, 2025, https://csrc.nist.gov/csrc/media/Presentations/2022/high-performance-hardware-implementation-of-lattic/images-media/session-4-beckwith-high-performance-hardware-pqc2022.pdf
19. arXiv:2211.12265v2 [cs.CR] 22 Apr 2023, accessed October 27, 2025, https://arxiv.org/pdf/2211.12265
20. High-Speed NTT-based Polynomial Multiplication Accelerator for Post-Quantum Cryptography | NSF Public Access Repository, accessed October 27, 2025, https://par.nsf.gov/biblio/10337501-high-speed-ntt-based-polynomial-multiplication-accelerator-post-quantum-cryptography
21. arxiv.org, accessed October 27, 2025, https://arxiv.org/html/2505.01782v1
22. Efficient unified architecture for post-quantum cryptography: combining Dilithium and Kyber, accessed October 27, 2025, https://peerj.com/articles/cs-2746/
23. MojoBench: Language Modeling and Benchmarks for Mojo - ACL Anthology, accessed October 27, 2025, https://aclanthology.org/2025.findings-naacl.230/
24. Mojo is Fast. But is it the Future of Python or Just a Flicker in the Dark? - Medium, accessed October 27, 2025, https://medium.com/@tfmv/mojo-is-fast-but-is-it-the-future-of-python-or-just-a-flicker-in-the-dark-3fe99ac83f5e

25. lib-Q — Rust crypto library // Lib.rs, accessed October 27, 2025, https://lib.rs/crates/lib-q
26. Framework for Prototyping And In-Hardware Evaluation of Post-Quantum Cryptography HW Accelerators (TU Darmstadt) - Semiconductor Engineering, accessed October 27, 2025, https://semiengineering.com/framework-for-prototyping-and-in-hardware-evaluation-of-post-quantum-cryptography-hw-accelerators-tu-darmstadt/
27. Future Proofing TLS1.3: Integration and Evaluation of Post-Quantum ..., accessed October 27, 2025, https://repository.tugraz.at/publications/marc21/0d0ft-bjs46/files/77392.pdf?download=1
28. cuML-DSA: Optimized Signing Procedure and Server-Oriented GPU Design for ML-DSA, accessed October 27, 2025, https://www.computer.org/csdl/journal/tq/2025/03/10748358/21HRyiBmEsU
29. [Blog] Setting the Standard: Industry First PQC-Ready FPGA - Lattice Semiconductor, accessed October 27, 2025, https://www.latticesemi.com/en/Blog/2025/10/09/14/54/Setting-the-Standard-Industry-First-PQC-Ready-FPGA
30. Quantum Shield for AI: Lattice Semiconductor Unveils Post-Quantum Secure FPGAs | User | chroniclejournal.com, accessed October 27, 2025, https://markets.chroniclejournal.com/chroniclejournal/article/tokenring-2025-10-14-quantum-shield-for-ai-lattice-semiconductor-unveils-post-quantum-secure-fpgas
31. European Payments Council's Guidelines on cryptographic algorithms usage and key management - EPC Document, accessed October 27, 2025, https://www.europeanpaymentscouncil.eu/sites/default/files/kb/file/2024-03/EPC342-08%20v13.0%20Guidelines%20on%20Cryptographic%20Algorithms%20Usage%20and%20Key%20Management.pdf
32. NTT and Inverse NTT Quantum Circuits in CRYSTALS-Kyber for Post-Quantum Security Evaluation - ResearchGate, accessed October 27, 2025, https://www.researchgate.net/publication/373996716_NTT_and_Inverse_NTT_Quantum_Circuits_in_CRYSTALS-Kyber_for_Post-Quantum_Security_Evaluation
33. Evaluating the security of CRYSTALS-Dilithium in the quantum random oracle model - National Institute of Standards and Technology, accessed October 27, 2025, https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=956883
34. Decrypting the NIST-Approved Algorithms for Enterprises | Encryption Consulting, accessed October 27, 2025, https://www.encryptionconsulting.com/decrypting-the-nist-approved-algorithms-for-enterprises/
35. Performance and Storage Analysis of CRYSTALS-Kyber (ML-KEM) as a Post-Quantum Replacement for RSA and ECC - arXiv, accessed October 27, 2025, https://arxiv.org/html/2508.01694v4
36. Cryptographic Agility for Real-Time Payment Systems: A Post- Quantum Architecture and Performance Evaluation, accessed October 27, 2025, https://jisem-journal.com/index.php/journal/article/download/13171/6155/22225

37. Session VII – NIST 6 th PQC Standardization Conference NIST ..., accessed October 27, 2025, https://csrc.nist.gov/csrc/media/presentations/2025/nist-pqc-migration-project-and-crypto-agility-proj/nist_pqc_migration_project-newhouse_2.10.pdf
38. Preparing for Post-Quantum Critical Infrastructure: Assessments of Quantum Computing Vulnerabilities of National Critical Functi - RAND, accessed October 27, 2025, https://www.rand.org/content/dam/rand/pubs/research_reports/RRA1300/RRA1367-6/RAND_RRA1367-6.pdf
39. CNSA 2.0 Compliance Requirements, Algorithms & Timelines ..., accessed October 27, 2025, https://www.safelogic.com/compliance/cnsa-2
40. The Commercial National Security Algorithm Suite 2.0 and ... - DoD, accessed October 27, 2025, https://media.defense.gov/2022/Sep/07/2003071836/-1/-1/0/CSI_CNSA_2.0_FAQ_.PDF
41. Open Quantum Safe: Home, accessed October 27, 2025, https://openquantumsafe.org/
42. About our project | Open Quantum Safe, accessed October 27, 2025, https://openquantumsafe.org/about/
43. Demo using Bouncy Castle Post Quantum Cryptography Library in Android Kotlin Project (NTRU and AES-128) - GitHub, accessed October 27, 2025, https://github.com/snowfluke/bouncy-castle-pqc-android-kotlin-demo
44. EJBCA and post quantum preparedness | KvantPhone, accessed October 27, 2025, https://kvantphone.com/en/ejbca-and-post-quantum-preparedness/
45. Support - ID Quantique, accessed October 27, 2025, https://www.idquantique.com/support/
46. A Compact and High-Performance Hardware Architecture for CRYSTALS-Dilithium, accessed October 27, 2025, https://www.researchgate.net/publication/356399524_A_Compact_and_High-Performance_Hardware_Architecture_for_CRYSTALS-Dilithium
47. Announcing the Commercial National Security Algorithm Suite 2.0, accessed October 27, 2025, https://media.defense.gov/2025/May/30/2003728741/-1/-1/0/CSA_CNSA_2.0_ALGORITHMS.PDF
48. FIPS 203, Module-Lattice-Based Key-Encapsulation Mechanism ..., accessed October 27, 2025, https://csrc.nist.gov/pubs/fips/203/final